

Practical no. 4

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Class :- CO3K – B

Subject :- DSU

Program :-

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int arr[10],key;
    int n,i,mid,high,low;
    clrscr();
    printf("Enter no. of elements :- ");
    scanf("%d",&n);
    printf("\nEnter %d elements :- \n",n);
    for(i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    printf("\nEnter element to be searched :- ");
    scanf("%d",&key);
    //initialising low and high elements
    low=0;
    high=n-1;
    while(low<=high)
    {
        mid=(low+high)/2;
        if(arr[mid]==key)
        {
            printf("%d found at index %d",key,mid);
```

```
break;
}
else if(arr[mid]<key)
{
low=mid+1;
}
else
{
high=mid-1;
}
}
if(low>high)
{
printf("%d is not present in the list",key);
}
getch();
}
```

OUTPUT :-

```
NeuTroN DOS-C++ 0.77, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
Enter no. of elements :- 5

Enter 5 elements :-
10 20 30 40 50

Enter element to be searched :- 40
40 found at index 3
```

```
NeuTroN DOS-C++ 0.77, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
Enter no. of elements :- 5

Enter 5 elements :-
10 20 30 40 50

Enter element to be searched :- 60
60 is not present in the list
```